

Effects of Herbal Medicinal Products and Food Supplements on Induction of CYP1A2, CYP3A4 and MDR1 in the Human Colon Carcinoma Cell Line LS180

Helena Brandin, Eila Viitanen, Olle Myrberg and Ann-Kristin Arvidsson*

Medical Products Agency, Uppsala, Sweden

A selection of popular herbal medicinal products and food supplements were analysed for their potential to modulate the expression of the cytochrome P450 enzymes CYP1A2 and CYP3A4 and the transporter protein MDR1. A total of 31 products were analysed. Nine of the products have been approved by the Medical Products Agency (MPA) in Sweden and are marketed as herbal medicinal products. Twenty-two of the products have not been assessed by the MPA and are marketed as food supplements. LS180 cells were exposed to extracts from the different herbal products and real-time quantitative polymerase chain reaction, RT-QPCR, was subsequently used to analyse the relative mRNA levels of CYP1A2, CYP3A4 or MDR1 in treated and non-treated cells. Our results show that 17 of 31 products tested induced a two-fold expression or more for at least one of the genes analysed. Four products, of which a ginger-supplement was the most potent, induced all three genes. Copyright © 2006 John Wiley & Sons, Ltd.

Keywords: herbal medicinal products; cytochrome P450 enzymes; induction; real-time quantitative polymerase chain reaction.

INTRODUCTION

The chemical composition and pharmacological properties of many herbal medicines are not always well known. Many of the herbal products available over the counter today have not been assessed by regulatory authorities and they are often promoted as safe products with no side effects. Some of the constituents in herbal products may be substrates, inhibitors or inducers of the cytochrome P450 enzymes (CYPs), and thereby have an impact on the pharmacokinetics of a coadministered drug metabolized by the CYP enzymes. The most well-known example of a drug–herb interaction is St John's Wort. Several reports have shown an increased clearance of CYP3A4 substrates such as indinavir, cyclosporine and oral contraceptives, in patients concomitantly taking St John's wort (Piscitelli *et al.*, 2000; Ahmed *et al.*, 2001). The CYP3A4 enzyme is the most abundantly expressed CYP in the human liver and small intestine and metabolizes more than 50% of all marketed pharmaceuticals. Induction of CYP1A2 can cause adverse effects such as enhanced metabolic activation of carcinogenic aromatic amines and polycyclic aromatic carbons, as well as increased metabolic clearance of drugs (McManus *et al.*, 1990). CYP1A2 is involved in the metabolism of common drugs such as caffeine and theophylline (Campbell *et al.*, 1987). MDR1 is the gene encoding p-glycoprotein (p-gp), which is one of the most important membrane-

bound drug transporters involved in the systemic disposition of drugs. P-gp shares many substrates with CYP3A4 and the regulation of the two genes partly involves the same set of transcription factors (Geick *et al.*, 2001). Induction of the genes for MDR1 and CYP3A4 could thus result in a reduced uptake as well as an increased metabolic clearance of a particular drug.

LS180 is a human colon carcinoma cell line that has been shown to be suitable for induction studies *in vitro*. The LS180 cells express many of the CYP isoforms, including CYP1A2 and CYP3A4, and have also retained the inducibility of these CYP isoforms (Engman *et al.*, 2001; Brandon *et al.*, 2006). Real time PCR (RT-PCR) is one of the most sensitive and accurate methods for quantifying mRNA expression levels (Wang and Brown, 1999). The method allows a comparison between expression levels of genes in different samples when normalization is performed using an endogenous reference gene. The aim of this study was to analyse a selection of popular herbal medicinal products and food supplements for their potential to induce the expression of the cytochrome P450 enzymes CYP1A2 and CYP3A4 and the transporter protein MDR1.

MATERIALS AND METHODS

Chemicals. LS180 cells were obtained from American Type Culture Collection (Manassas, VA, USA). Minimal essential medium (MEM) alpha medium, fetal bovine serum, penicillin and L-glutamate were obtained from Invitrogen (Carlsbad, CA, USA). Rifampicin and DMSO were obtained from Sigma Aldrich (St Louis,

* Correspondence to: Ann-Kristin Arvidsson, Medical Products Agency, Laboratory Department, Box 26, 751 03 Uppsala, Sweden.
E-mail: ann-kristin.arvidsson@mpa.se

MO, USA). Omeprazole (Losec 40 mg, AstraZeneca, Södertälje, Sweden) was obtained from the local pharmacy. Herbal medicinal products and food supplements were obtained from local pharmacies and health shops (Table 1).

Extract preparation. The amount given as the recommended daily dose (approximately 300–400 mg for most of the products) was used as the starting material for the extraction. The product was extracted with 50 mL 60% (v/v) ethanol in an ultrasound bath for 15 min followed by filtration and evaporation of the solvent. The dry extract was weighed and resolved in DMSO to give a concentration of 100 mg/mL. The DMSO extract was then diluted with MEM and added to the cells at a final concentration of 100 µg/mL. Products already distributed as liquid extract were diluted 1:500 with MEM.

Cell culture and extract treatment. LS180 cells were maintained in a humidified 37 °C incubator with a 5% carbon dioxide in air atmosphere in MEM alpha medium supplemented with 10% fetal calf serum, 2% L-glutamate 1% penicillin/streptomycin. Rifampicin was dissolved in DMSO and added to the cells at a final concentration of 50 µM. The final DMSO concentration in the medium was 0.1%. Omeprazole was dissolved in water (20 mM), diluted in MEM and added to the cells at a final concentration of 100 µM. Control cells were treated with vehicle alone. For induction studies the cells were seeded onto 24-well plates 24 h before adding extract or reference substances and was 80%–90% confluent at the start of the experiment. Incubation was performed in 2 or 4 replicate wells for each product, and cells were exposed for 48 h with a daily medium change.

Table 1. List of constituents, trade names and manufacturers/distributors for products used in the present study

Main constituent(s)	Trade name (abbreviation)	Manufacturer/Distributor
<i>AloeVera Barbadensis</i>	AloeVera Juice (AA)	K & L Labs, CA, USA
Arctic root (<i>Rhodiola rosea</i>)	Extra stark Rosenrot (RT)	Örtmedicinska Institutet, Malmö, Sweden
<i>Arctostaphylos uva-ursi</i>	Uva-E ^a (UE)	Pharbio Medical Int. AB, Göteborg, Sweden
<i>Astragalus membranaceus</i>	Huang qi (AS)	Ginseng.se
Bitter root (<i>Gentiana lutea</i>)	Sinova ^a (SA)	Boehringer Ingelheim AB, Skärholmen, Sweden
Chinese Hawthorn Fruit	Shan Zha (KH)	Ginseng.se
Chrysanthemum	Vild Ju Hua (CM)	Ginseng.se
Fish oil mixture/evening primrose oil	Efalex (EF)	Efamol Ltd, Selby, England
Ginger (<i>Zingiber officinale</i> , <i>Alpinia galanga</i> root)	Zinaxin (ZN)	Ferrosan A/S Soeborg, Denmark
Grapefruit seed extract	Citroseed (CI)	Cintamani Manufacturing AB, Eslöv, Sweden
Herbal tincture	Alpenkraft (AT)	SalusHaus D-Bruckmuhl-Obb., Germany
Herbal yeast extract ((<i>Saccharomyces cerevisiae</i>) + vitamins minerals etc)	Bio-Strath (BS)	Bio-Strath AG, Zurich, Switzerland
Horse chestnut (<i>Aesculus hippocastanum</i>)	Venastat ^a (VT)	Boehringer Ingelheim AB, Skärholmen, Sweden
Linseed oil	Linfröolja (LA)	Helhetshälsa, Borghamn, Sweden
<i>Lonicera japonica</i>	Jin yin hua (LJ)	Ginseng.se
<i>Lycium barbarum</i>	Gou gi zi (LB)	Ginseng.se
New Zealand green lipped mussel extract (<i>Perna canaliculus</i>)	Lyprinol (LL)	Pharmlink Ltd, Burleigh Waters, Australia
Oxerutin, Zinkgluconat	Coldizin (CO)	ProMedic Naturläkemedel AB, Huskvarna, Sweden
Panax ginseng	Red ginseng (RG)	Ginseng.se
Panax ginseng	White ginseng (VG)	Ginseng.se
Panax ginseng mix	Panax ginseng (GP)	Ginseng.se
Pollen extract	Cernitol ^a (CL)	Cederroth International AB, Upplands Väsby, Sweden
Pollen extract	Femal Balans (FB)	Natumin Pharma AB, Huskvarna, Sweden
Rosehip (Hyben Vital Lito)	Litozin (LN)	Dansk Droge, Ishøj, Denmark
<i>Salvia officinalis</i>	NoSweat ^a (NS)	Biokraft Pharma AB, Helsingborg, Sweden
Saw Palmetto (<i>Serenoa repens</i>)	Sabamin ^a (SA)	Svenska Bioforce AB, Lund, Sweden
<i>Schisandra chinensis</i>	Wu wei zi (SC)	Ginseng.se
<i>Schisandra chinensis</i> , <i>Eleuterococcus sentiosus</i> , <i>Rhodiola rosea</i>	Chisan adaptogen (CA)	Örtmedicinska Institutet, Malmö, Sweden
Siberian ginseng (<i>Eleuterococcus sentiosus</i>), <i>Schisandra chinensis</i>	Ryskrot/Schisandra (RY)	Örtmedicinska Institutet, Malmö, Sweden
<i>Solidago virgaurea</i>	Solidamin ^a (SO)	Bioforce, Roggwil, Switzerland
<i>Taraxacum officinale</i> , <i>Peumus boldus</i> , <i>Cynara scolymus</i> , <i>Mentha piperita</i>	Cynaramin ^a (CN)	Bioforce, Roggwil, Switzerland
<i>Valeriana officinalis</i>	Valeriana Forte ^a (VF)	Pharbio Medical Int. AB, Göteborg, Sweden
Reference substances		
Rifampicin	(RIF)	Sigma Aldrich, St Louis, MO, USA
Omeprazole	(OE)	Losec 40 mg, AstraZeneca, Södertälje, Sweden

^a Products have been approved by the Medical Products Agency, Sweden.

RNA extraction and RT-PCR. Total RNA from the LS180 cells was extracted using the RNEasy mini kit (Qiagen, Hilden, Germany), followed by DNase treatment (RNase free DNase I, Qiagen, Hilden, Germany). RNA quantity and quality of the different samples were determined by the 260/280 nm absorbance ratio using a Nanodrop 3000 (Nanodrop Technologies, Wilmington, DE, USA). 0.5 µg total RNA was used for cDNA synthesis using the iScript cDNA Synthesis Kit (Bio-Rad Laboratories, Hercules, CA, USA). Two cDNAs per purified RNA sample (herbal extract, inducer or vehicle) were synthesized. Running duplicates in the RT-step instead of the PCR-step reduces the total variation of final PCR-product since variation in the RT-step is usually larger than in the PCR-step (Ståhlberg *et al.*, 2004). Quantitative PCR was performed using gene specific primers CYP3A4; (forward 5'-GAAACACAGATCCCCCTGAA-3', reverse 5'-CCCGGTTATTTATGCAGTCC-3'; product size 216 bp), CYP1A2; (forward 5'-GTTCAAGCACAGCAAGAAGG-3, reverse 5'-CTCAGGCTTGGTCAAAAGGT-3'; product size 153 bp.) and MDR1; (5'-CTGTATTGTTTGCCACCACGA-3', reverse 5'-AGGGTGTCAAATTTATGAGGCAGT-3' product size 139 bp). The reference gene hypoxanthine ribosyltransferase, HPRT1 (forward 5'-TGACACTGGCAAACAATGCA-3', reverse 5'-GGTCCTTTTTCACCAAGCAAGCT-3'; product size 94 bp) was used for normalization of CYP1A2, CYP3A4 and MDR1 mRNA expression to enable cross-comparison among different samples. All primers were custom synthesized by Thermo Electron, Ulm, Germany. All primers were designed to span intron/exon borders in order to avoid amplifying possible genomic contamination. Real-time PCR and quantitation were performed using a Rotor-Gene 3000 (Corbett Research, Mortlake, Australia). All reactions were performed in a total of 20 µL containing 1x iQ SYBR Green Supermix (Bio-Rad Laboratories, Hercules, CA, USA), 250 nm of each primer and 2 µL cDNA (of 20 µL total) as template. PCR conditions were as follows: 95 °C for 3 min to activate the iTaq-polymerase, 40 cycles of 30 s at 95 °C, 30 s at 60 °C and 30 s at 72 °C. Melt curve analyses were performed for all samples and only sharp melting points were observed, indicating a specific signal and no primer dimers or mispriming. No template controls (NTC) were always included in each run.

Data analysis. For calculations and statistical comparison between the different treatment groups (treatment vs control), the randomized group-wise comparison REST©-software was used. The mathematical model used is based on the correction for exact PCR efficiencies and the mean crossing point deviation between sample group(s) and control group(s) (Pfaffl *et al.*, 2002). The number of samples per group was 4–8 and the total number of reallocations performed were $n = 2000$.

RESULTS

DMSO extracts from different herbal medicines and food supplements were tested for their potential to induce mRNA expression of CYP1A2, CYP3A4 and MDR1. The relative levels of mRNA were analysed after

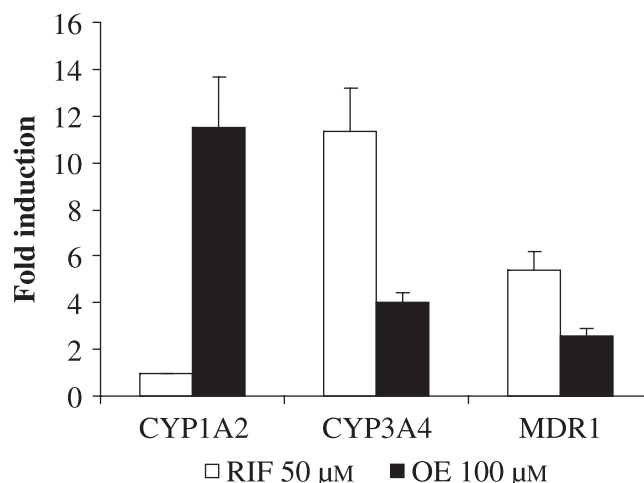


Figure 1. LS180 cells were exposed to 50 µM rifampicin (RIF) or 100 µM omeprazole (OE) for 48 h. mRNA levels were measured with RT-QPCR and normalized to HPRT1. Results are presented as mean significant fold induction ($\pm$ SE) of mRNA levels compared with vehicle control, $n = 4$.

48 h exposure of the LS180 cells to the DMSO extracts using RT-QPCR. Inducibility of CYP1A2, CYP3A4 and MDR1 in the LS180 cell line was shown by exposing the cells to well known inducers. Exposure of the cells to rifampicin induced expression of CYP3A4 and MDR1 mRNA, but did not significantly affect CYP1A2 mRNA levels. Omeprazole caused induction of CYP1A2 mRNA and also, to a lesser extent, induced CYP3A4 and MDR1 mRNA (Fig. 1). The LS180 cells were exposed to the inducers rifampicin and omeprazole in several different experiments to test the reproducibility of the LS180 cells as a model system for induction studies. The fold induction obtained was in the same range in all experiments and for all investigated mRNAs (data not shown).

The fold-induction obtained for each mRNA and herbal product is presented in Table 2. The results are shown as a significant fold-induction or no significant induction (NSI). Our results show that 55% of the analysed products caused a fold-induction of two or more for at least one of the mRNAs CYP1A2, CYP3A4 or MDR1. The most frequently induced mRNA was CYP1A2, which was induced by 39% the herbal products, followed by MDR1 (32%) and the least induced mRNA was CYP3A4 (23%). Four products, BS, RY, SC and ZN induced all three genes (induction of two or more). A food supplement containing ginger, ZN, was found to be the most potent inducer of all products analysed and caused a fold-induction of 11.2 for CYP1A2, 6.4 for CYP3A4 and 5.4 for MDR1. An herbal yeast extract supplemented with various vitamins and minerals, BS, caused 6.4-fold induction of CYP1A2, a 2.1-fold induction of CYP3A4 and a 2.5-fold induction of MDR1. Three different Schisandra-containing supplements, CA, RY and SC caused induction of both CYP enzymes as well as of MDR1. Extracts of arctic root, RT, fish oil/evening primrose, EF and horse chestnut, VT caused induction of only CYP1A2, but had no effect on the expression of CYP3A4 or MDR1. Some products, such as a sage-based product, NS, and Chinese hawthorne, KH, exclusively modulated expression of CYP3A4 and MDR1.

Table 2. Fold-induction of CYP1A2, CYP3A4 and MDR1 in LS180 cells after incubation with DMSO extracts from each product for 48 h

Product	CYP1A2	CYP3A4	MDR1
AloeVera Juice ^a (AA)	1.7 ± 0.3	1.8 ± 0.4	1.3 ± 0.3
Alpenkraft (AT)	NSI	1.5 ± 0.2	NSI
Bio-Strath ^a (BS)	6.4 ± 1.3	2.1 ± 0.3	2.5 ± 0.3
Cernitol ^a (CL)	4.4 ± 0.8	1.7 ± 0.2	1.7 ± 0.2
Chisan adaptogen ^a (CA)	1.7 ± 0.3	2.0 ± 0.2	1.8 ± 0.1
Citroseed (CI)	NSI	NSI	NSI
Coldizin (CO)	NSI	NSI	2.1 ± 0.2
Cynaramin (CN)	NSI	NSI	5.4 ± 0.9
Efalex ^a (EF)	6.1 ± 1.7	NSI	NSI
Extra stark Rosenrot ^a (RT)	6.3 ± 2.0	NSI	NSI
Femal Balans (FB)	2.4 ± 0.2	NSI	4.3 ± 0.4
Gou gi zi (LB)	NSI	NSI	NSI
Huang qi (AS)	NSI	NSI	NSI
Jin yin hua (LJ)	1.7 ± 0.3	NSI	NSI
Linfröolja (LA)	NSI	NSI	NSI
Lyprinol (LL)	NSI	NSI	NSI
NoSweat (NS)	NSI	2.2 ± 0.5	4.8 ± 0.6
Panax ginseng (GP)	NSI	NSI	NSI
Red ginseng (RG)	NSI	NSI	NSI
Rysk rot/Schisandra ^a (RY)	2.3 ± 0.4	2.3 ± 0.2	2.1 ± 0.1
Sabamin (SN)	1.7 ± 0.4	1.2 ± 0.1	NSI
Shan Zha (KH)	NSI	2.5 ± 0.5	2.4 ± 0.3
Sinova (SA)	2.1 ± 0.3	NSI	4.0 ± 0.4
Solidamin (SO)	NSI	1.9 ± 0.2	NSI
Uva-E ^a (UE)	2.7 ± 0.5	1.5 ± 0.2	1.8 ± 0.3
Valeriana Forte (VF)	NSI	NSI	NSI
Venastat (VT)	2.6 ± 0.6	NSI	NSI
White ginseng (VG)	NSI	NSI	NSI
Vild Ju Hua (CM)	2.2 ± 0.2	NSI	1.7 ± 0.2
Wu wei zi ^a (SC)	3.9 ± 0.7	3.3 ± 0.4	4.5 ± 0.6
Zinaxin ^a (ZN)	11.2 ± 2.8	6.4 ± 0.9	5.4 ± 0.9

All mRNA data were normalized using the reference gene HPRT1. Fold-induction were calculated using software program REST[®] which uses randomized group-wise comparison for statistical comparison between treatment and control groups. The number of samples for each group were 4 except for products^a where $n = 8$, and the number of reallocations performed were 2000. NSI, no significant induction. Results are presented $\pm$ SE, $p < 0.05$.

DISCUSSION

Reliable methodology that allows one to evaluate the induction of human CYPs by new molecular entities is very important. Since induction of both CYP1A2 and 3A4 show significant species specificity the use of human systems is required (Vignati *et al.*, 2004; Backlund and Ingelman-Sundberg, 2004). Primary cultures of human hepatocytes would thus be the ideal experimental model for evaluating induction of CYP enzymes. However, due to the difficulty in obtaining liver samples and the significant variability among samples there is a need for alternative *in vitro* systems. Reporter gene systems are also widely used but have the drawback of having a narrow specificity. In this study, the LS180 cell line was used as a tool for assessing CYP-induction. The LS180 cells express many of the CYP isoforms, including CYP1A2 and CYP3A4, which allows the assessment of induction of several CYP enzymes in the same experiment (Engman *et al.*, 2001; Brandon *et al.*, 2006).

Induction of cytochrome P450 genes generally occurs at the transcriptional level and is mediated by certain receptors that act as transcription factors. The nuclear

receptor PXR is believed to be the master regulator of CYP3A4 gene expression. Another means of CYP3A4 induction is via CAR-activation. CAR and PXR regulate an overlapping but not identical set of genes, and many compounds can activate both receptors (Moore *et al.*, 2000). Binding of an agonistic compound is generally accepted as the activation mechanism for PXR (Ekins and Erickson, 2002). The activation of CAR is more complex and difficult to study since transformed cells seem to have lost the capability to retain CAR in the cytoplasm and thereby CAR functions as a constitutively active receptor in *in vitro* cell-based assays (Honkakoski *et al.*, 2003). The MDR1 gene encodes p-gp which is one of the most important membrane-bound drug transporters and is highly expressed in the small intestine. Induction of p-gp expression should also be considered when assessing CYP3A4 induction since both proteins can contribute to the overall clearance of a compound. The mechanism of CYP1A2 induction has been well studied and involves ligand-mediated activation of the AhR (Withlock *et al.*, 1999; Black and Quattrochi, 2004). PXR, CAR and AhR are expressed in both hepatocytes and intestinal cells *in vivo* and are involved in regulation of the expression of CYP3A4/MDR1 and CYP1A2

in both cell types (Nebert *et al.*, 2004; Handschin and Meyer, 2003). However, discrepancy between hepatocyte and intestinal regulation is possible and has also been suggested for CYP3A4 (Gorski *et al.*, 2004; Hartley *et al.*, 2004). The LS180 cells express all of the above mentioned receptors and therefore represent an appropriate cellular context for the study of CYP1A2-, CYP3A4- and MDR1-induction (Harper *et al.*, 1991; Prufunder *et al.*, 2003).

The herbal products analysed in this study contain complex mixtures of phytochemicals that potentially could act as inducers of drug-metabolizing enzymes. Some of the analysed products are mixtures of several herbs and some also contain vitamins and minerals. This makes it difficult to identify the active substance(s) that could be responsible for the induction caused by these products. Furthermore, the concentration of the active substance(s) in the different DMSO extracts might not be clinically relevant. The bioavailability of the herbal medicine will also affect the overall clearance *in vivo*.

Interestingly, in a recent study Mu and coworkers identified two constituents of the traditional Chinese herb Schisandra, schisandrol and schisandrin that induce the expression of CYP3A isozymes, both in a PXR reporter gene system and in primary hepatocyte cultures (Mu *et al.*, 2006). This is in line with the present results in which it was found that exposure of the LS180 cells to extracts of three different Schisandra-products caused induction of CYP3A4 and MDR1. Additional studies are required to elucidate the mechanism behind the observed induction but it seems likely that the increased expression of CYP3A4 mRNA that is seen in the LS180 cells could be due to activation of PXR,

although a contribution of CAR mediated induction cannot be ruled out. Flavonoids are a group of structurally related phytochemicals that exhibit a wide range of biological activities arising mainly from their antioxidant properties. Some flavonoids such as flavone, tangeretin and the synthetic β -naphthoflavone have been reported to stimulate the expression of CYP1A1/2 (Zhou *et al.*, 2003). This is also in accordance with our results that found induction of CYP1A2 by products made from flavonoid-rich herbs such as Horse chestnut and Arctic root. The observed induction of CYP1A2, CYP3A4 and MDR1 by the ginger supplement is remarkable. To our knowledge this effect of ginger has not been previously reported in a human *in vitro* cell system. The true risk of the observed effect is difficult to predict without further *in vivo* studies, however, it could be of clinical significance if ginger supplements are taken together with drugs which are substrates for CYP1A2 or CYP3A4.

In conclusion, the LS180 cell line was used as an experimental model system to study the effect of herbal medicines and food supplements on the induction of CYP1A2, CYP3A4 and MDR1. Seventeen of 31 products tested were found to cause an induction of two or more for at least one of the analysed mRNAs. CYP1A2 was induced by 39% of the products while CYP3A4 and MDR1 were induced by 23% and 32%, respectively. The results suggest that there is a risk of potential herbal medicine/food supplement–drug interaction. However, it should be recognized that the assessment of the induction potential of a compound *in vitro* is very difficult, therefore these findings require further *in vivo* studies.

REFERENCES

- Ahmed SM, Banner NR, Dubrey SW. 2001. Low cyclosporin-A level due to Saint-John's-wort in heart transplant patients. *J Heart Lung Transplant* **20**: 795.
- Backlund M, Ingelman-Sundberg M. 2004. Different structural requirements of the ligand binding domain of the aryl hydrocarbon receptor for high- and low-affinity ligand binding and receptor activation. *Mol Pharmacol* **65**: 416–425.
- Black VH, Quattrochi LC. 2004. Molecular cloning of the guinea pig CYP1A2 gene 5' flanking region: identification of functional aromatic hydrocarbon response element and characterization of CYP1A2 expression in GPC16 cells. *Drug Metab Dispos* **32**: 595–602.
- Brandon EFA, Bosch TM, Deenen MJ *et al.* 2006. Validation of *in vitro* models used in drug metabolism and transport studies; genotyping of cytochrome P450, phase II enzymes and drug transporter polymorphism in the human hepatoma (HepG2), ovarian carcinoma (IGROV-1) and colon carcinoma (CaCo-2, LS180) cell lines. *Toxicol Appl Pharmacol* **211**: 1–10.
- Campbell ME, Grant DM, Inaba T, Kalow W. 1987. Biotransformation of caffeine, paraxanthine, theophylline and theobromine by polycyclic aromatic inducible cytochrome(s) P-450 in human liver microsomes. *Drug Metab Dispos* **15**: 237–249.
- Ekins S, Erickson JA. 2002. A pharmacophore for human pregnane X receptor ligands. *Drug Metab Dispos* **30**: 96–99.
- Engman HA, Lennernäs H, Taipalensuu J, Otter C, Leidvik B, Artursson P. 2001. CYP3A4, CYP3A5, and MDR1 in human small and large intestinal cell lines suitable for drug transport studies. *J Pharm Sci* **90**: 1736–1751.
- Geick A, Eichelbaum M, Burk O. 2001. Nuclear receptor response elements mediate induction of intestinal MDR1 by rifampin. *J Biol Chem* **276**: 14581–14587.
- Gorski JC, Huang SM, Pinto A, Hamman MA, Hilligoss JK, Zaheer NA, Desai M, Miller M, Hall SD. 2004. The effect of echinacea (Echinacea purpurea root) on cytochrome P450 activity *in vivo*. *Clin Pharmacol Ther* **75**: 89–100.
- Handschin C, Meyer UA. 2003. Induction of drug metabolism: The role of nuclear receptors. *Pharmacol Rev* **55**: 649–673.
- Harper PA, Prokipcak RD, Bush LE, Golas CL, Okey AB. 1991. Detection and characterization of the Ah receptor in the human colon adenocarcinoma cell line LS180. *Arch Biochem Biophys* **290**: 27–36.
- Hartley DP, Dai X, He YD, Carlini EJ, Wang B, Huskey SE, Ulrich RG, Rushmore TH, Evers R, Evans DC. 2004. Activators of the rat pregnane X receptor differentially modulate hepatic and intestinal gene expression. *Mol Pharmacol* **65**: 1159–1171.
- Honkakoski P, Sueyoshi T, Negishi M. 2003. Drug-activated nuclear receptors CAR and PXR. *Ann Med* **35**: 172–182.
- McManus ME, Burgess WM, Veronese ME, Huggett A, Quattrochi LC, Tuckey RH. 1990. Metabolism of 2-acetylaminofluorene and benzo(a)pyrene and activation of food-derived heterocyclic amine mutagens by human cytochromes P-450s. *Cancer Res* **50**: 3367–3376.
- Moore LM, Parks DJ, Jones SA *et al.* 2000. Orphan nuclear receptors constitutive androstane receptor and pregnane X receptor share xenobiotic and steroid ligands. *J Biol Chem* **275**: 15122–15127.
- Mu Y, Zhang J, Zhang S *et al.* 2006. Traditional Chinese medicines Wu Wei Zi (*Schisandra Chinesis* Baill) and Gan Cao (*Glycyrrhiza uralensis* Fisch) activate PXR and increase warfarin clearance in rats. *J Pharmacol Exp Ther* **316**: 1369–1377.
- Nebert DW, Dalton TP, Okey AB, Gonzalez FJ. 2004. Role of the aryl hydrocarbon receptor-mediated induction of the CYP1 enzymes in environmental toxicity and cancer. *J Biol Chem* **279**: 23847–23850.

- Pfaffl MW, Horgan GW, Dempfle L. 2002. Relative expression software tool (REST) for group-wise comparison and statistical analysis of relative expression results in real-time PCR. *Nucleic Acid Res* **30**: (9) e36.
- Piscitelli SC, Burstein AH, Chait D, Alfaro RM, Fallon J. 2000. Indinavir concentrations and St. John's wort. *Lancet* **355**: 547–548.
- Prufunder A, Gutmann H, Beglinger C, Drewe J. 2003. Gene expression of CYP3A4, ABC- transporters (MDR1 and MRP1-5) and PXR in three different human colon carcinoma cell lines. *J Pharm Pharmacol* **55**: 59–66.
- Ståhlberg A, Håkansson J, Xian X, Semb H, Kubista M. 2004. Properties of the reverse transcription reaction in mRNA quantification. *Clin Chem* **50**: 509–515.
- Vignati LA, Bogni A, Grossi P, Monshouwer M. 2004. A human and mouse pregnane X receptor reporter gene assay in combination with cytotoxicity measurements as a tool to evaluate species-specific CYP3A induction. *Toxicology* **199**: 23–33.
- Wang T, Brown MJ. 1999. mRNA quantification by real time TaqMan polymerase chain reaction: validation and comparison with RNase protection assay. *Anal Biochem* **269**: 201.
- Withlock Jr JP. 1999. Induction of cytochrome P4501A1. *Annu Rev Pharmacol Toxicol* **39**: 103–125.
- Zhou S, Gao Y, Jiang W, Huang M, Xu A, Paxton JW. 2003. Interactions of herbs with cytochrome P450. *Drug Metab Rev* **35**: 35–98.